Main project - Using ansible

Description

Instructions:

1. Install the necessary packages, which include MySQL 10.6, PHP version 8 or higher, and Nginx on your Linux server.
   2. Create a user's home directory under the '/home' directory. You should set the root directory for the website to be '/home/username/websitename/public.'
   3. Configure SFTP (Secure File Transfer Protocol) access for the user to ensure secure file management.
   4. Configure phpmyadmin access for the user to ensure secure database management.
   4. Install a free SSL certificate to enable secure HTTPS access to your website.
   5. Once the website is hosted, proceed to the WordPress dashboard to make necessary configurations and customizations.
   6. Write a personal blog post about yourself using the WordPress content management system.

Introduction

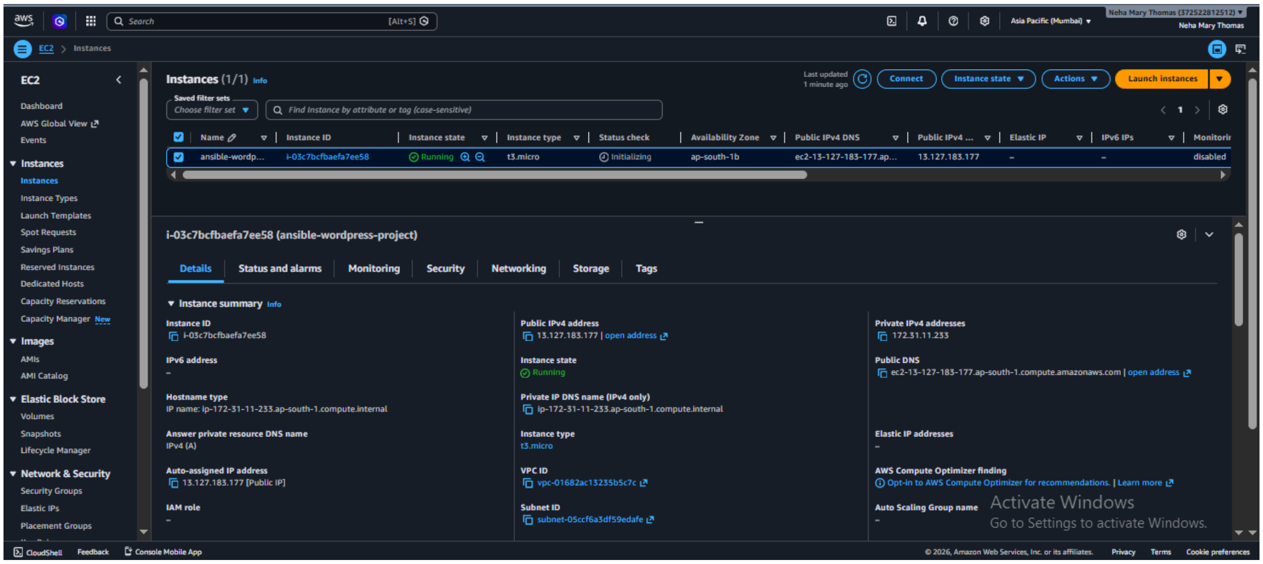
This project demonstrates the automated deployment and configuration of a WordPress-based website using Ansible on an AWS EC2 instance running Amazon Linux. The implementation includes the installation and configuration of Nginx, MySQL, PHP, WordPress, phpMyAdmin, SFTP access, domain and DNS configuration, and SSL certificate integration. The project highlights the use of infrastructure automation and web hosting technologies to create a secure, manageable, and scalable web environment.

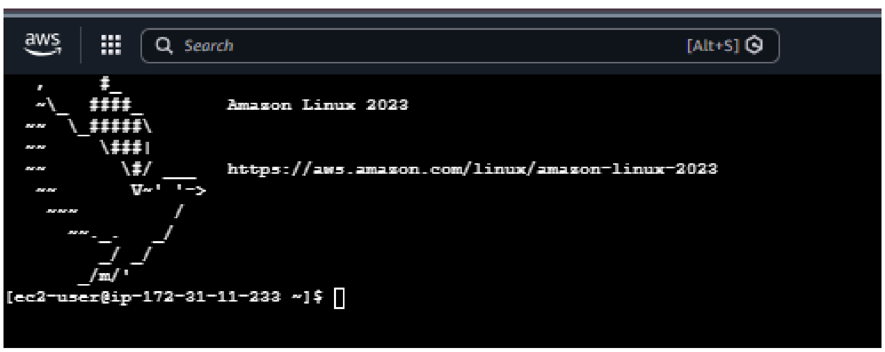
Objective

The primary objective of this project is to automate the deployment and configuration of a WordPress website using Ansible on an AWS cloud environment. The project aims to install and configure the required software components, create a secure user environment, enable database and file management through phpMyAdmin and SFTP, implement HTTPS security using SSL certificates, and successfully publish a personal blog through the WordPress Content Management System.

1. AWS EC2 Instance Creation

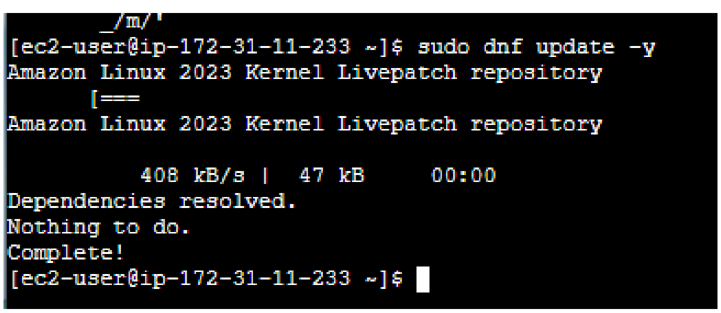
Successfully launched an Amazon Linux EC2 instance in AWS and connected through SSH.

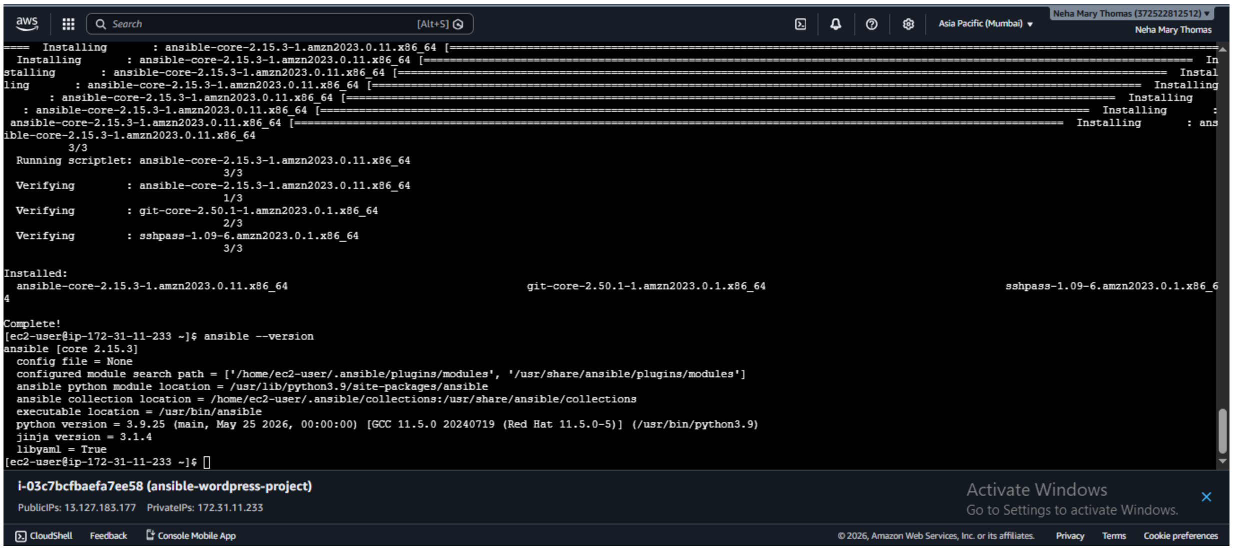




2. Installing Ansible

Installed and verified Ansible on the control node.





3. Creating Ansible Project Directory

A separate Ansible project workspace was successfully created, providing a centralized location for storing and managing all project automation files.

```
[ec2-user@ip-172-31-11-233 ~]$ mkdir ~/wordpress-project
[ec2-user@ip-172-31-11-233 ~]$ cd ~/wordpress-project
[ec2-user@ip-172-31-11-233 wordpress-project]$ pwd
/home/ec2-user/wordpress-project
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

4. Creating Ansible Inventory

Created inventory file for managing target hosts using Ansible.

```
[ec2-user@ip-172-31-11-233 wordpress-project]$ vim inventory
[ec2-user@ip-172-31-11-233 wordpress-project]$ cat inventory
[webserver]
localhost ansible_connection=local
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

5. Ansible Playbook Execution

Executed playbook to automate package installation and configuration.


[ec2-user@ip-172-31-11-233 wordpress-project]$ vim playbook.yml
[ec2-user@ip-172-31-11-233 wordpress-project]$ cat playbook.yml
---
- hosts: webserver
  become: yes

  tasks:

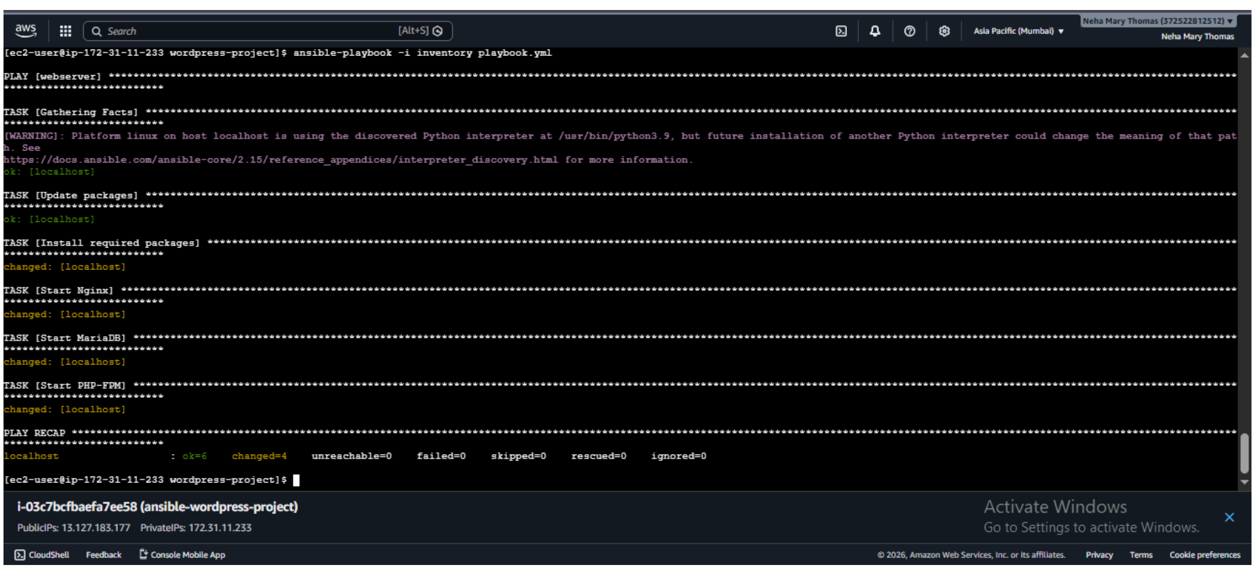
    - name: Update packages
      dnf:
        name: "*"
        state: latest

    - name: Install required packages
      dnf:
        name:
          - nginx
          - mariadb105-server
          - php
          - php-fpm
          - php-mysqlnd
          - php-gd
          - php-mbstring
          - php-xml
          - php-json
          - wget
          - unzip
          - tar
        state: present

    - name: Start Nginx
      service:
        name: nginx
        state: started
        enabled: yes

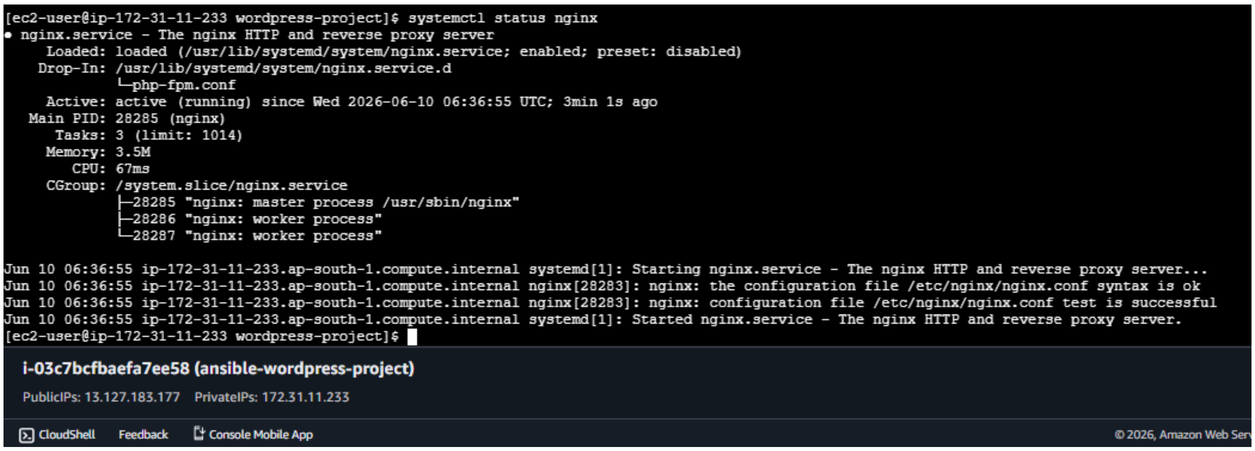
    - name: Start MariaDB
      service:
        name: mariadb
        state: started
        enabled: yes

    - name: Start PHP-FPM
      service:
        name: php-fpm
        state: started
        enabled: yes
[ec2-user@ip-172-31-11-233 wordpress-project]$
i-03c7bcfbaefa7ee58 (ansible-wordpress-project)
PublicIPs: 13.127.183.177 PrivateIPs: 172.31.11.233
CloudShell
Feedback
Console Mobile App

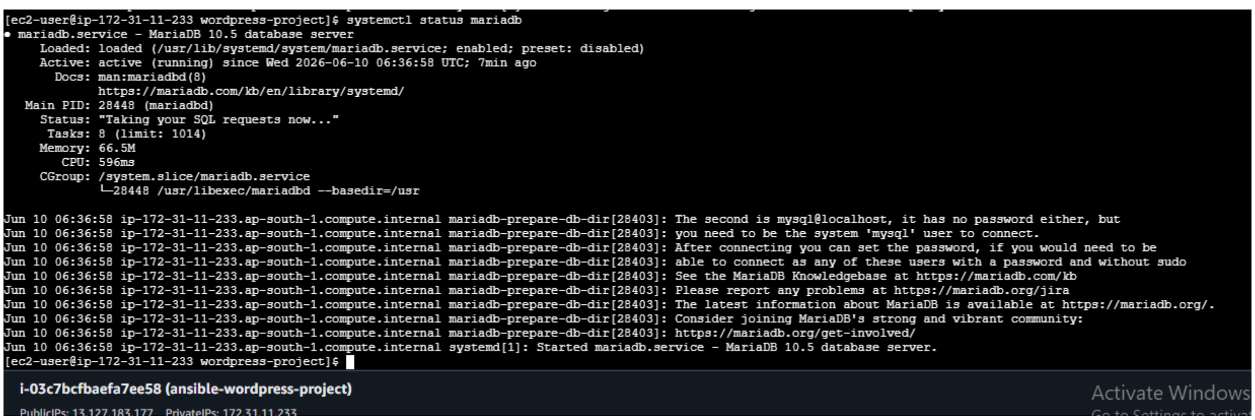

aws
Search
[Alt+S]
Asia Pacific (Mumbai)
Neha Mary Thomas
[ec2-user@ip-172-31-11-233 wordpress-project]$ ansible-playbook -i inventory playbook.yml
PLAY [webserver]
TASK [Gathering Facts]
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that pat
h. See
https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [localhost]
TASK [Update packages]
ok: [localhost]
TASK [Install required packages]
changed: [localhost]
TASK [Start Nginx]
changed: [localhost]
TASK [Start MariaDB]
changed: [localhost]
TASK [Start PHP-FPM]
changed: [localhost]
PLAY RECAP
localhost : ok=6 changed=4 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
[ec2-user@ip-172-31-11-233 wordpress-project]$
i-03c7bcfbaefa7ee58 (ansible-wordpress-project)
PublicIPs: 13.127.183.177 PrivateIPs: 172.31.11.233
Activate Windows
Go to Settings to activate Windows.
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

6. Verify Services

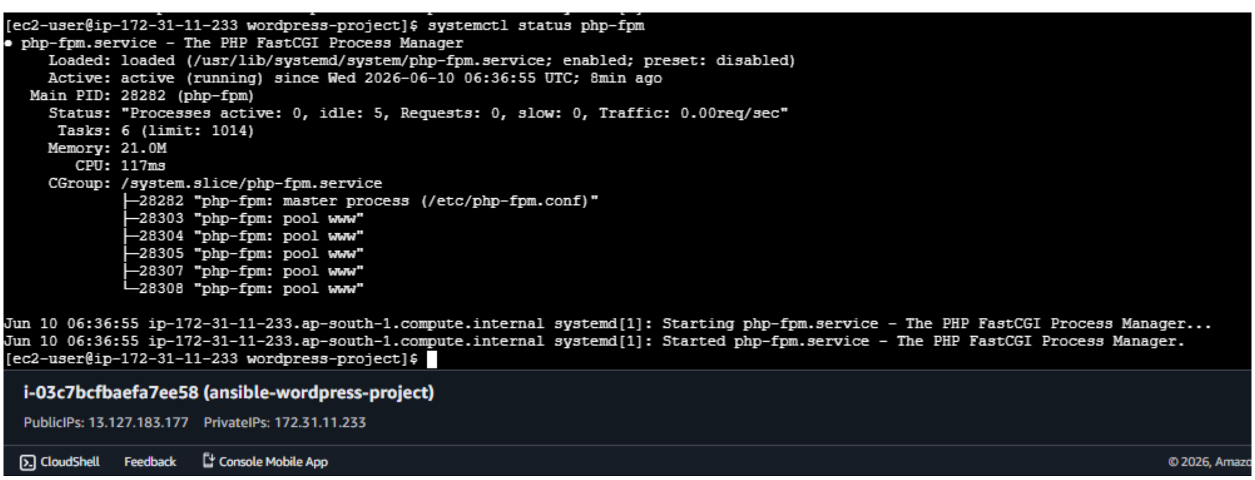
All services were found to be in the **active (running)** state, confirming that the server environment was correctly configured and operational.



```
[ec2-user@ip-172-31-11-233 wordpress-project]$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/nginx.service.d
             └─php-fpm.conf
     Active: active (running) since Wed 2026-06-10 06:36:55 UTC; 3min 1s ago
   Main PID: 28285 (nginx)
      Tasks: 3 (limit: 1014)
     Memory: 3.5M
        CPU: 67ms
     CGroup: /system.slice/nginx.service
             ├─28285 "nginx: master process /usr/sbin/nginx"
             ├─28286 "nginx: worker process"
             └─28287 "nginx: worker process"

Jun 10 06:36:55 ip-172-31-11-233.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jun 10 06:36:55 ip-172-31-11-233.ap-south-1.compute.internal nginx[28283]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jun 10 06:36:55 ip-172-31-11-233.ap-south-1.compute.internal nginx[28283]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jun 10 06:36:55 ip-172-31-11-233.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

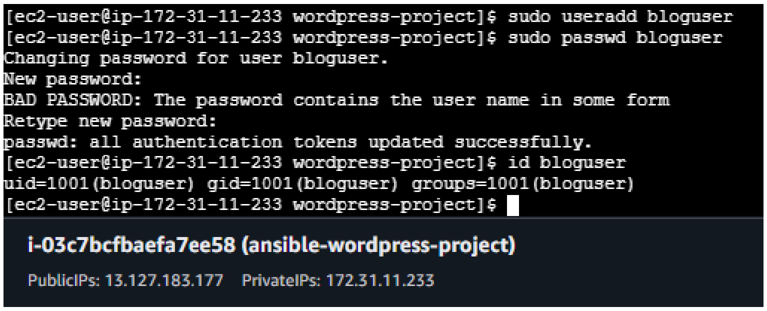
```
[ec2-user@ip-172-31-11-233 wordpress-project]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-06-10 06:36:58 UTC; 7min ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 28448 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 1014)
     Memory: 66.5M
        CPU: 596ms
     CGroup: /system.slice/mariadb.service
             └─28448 /usr/libexec/mariadbd --basedir=/usr

Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: The second is mysql@localhost, it has no password either, but
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: you need to be the system 'mysql' user to connect.
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: After connecting you can set the password, if you would need to be
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: able to connect as any of these users with a password and without sudo
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: Please report any problems at https://mariadb.org/jira
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: The latest information about MariaDB is available at https://mariadb.org/.
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: Consider joining MariaDB's strong and vibrant community:
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal mariadb-prepare-db-dir[28403]: https://mariadb.org/get-involved/
Jun 10 06:36:58 ip-172-31-11-233.ap-south-1.compute.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

```
[ec2-user@ip-172-31-11-233 wordpress-project]$ systemctl status php-fpm
● php-fpm.service - The PHP FastCGI Process Manager
     Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-06-10 06:36:55 UTC; 8min ago
   Main PID: 28282 (php-fpm)
     Status: "Processes active: 0, idle: 5, Requests: 0, slow: 0, Traffic: 0.00req/sec"
      Tasks: 6 (limit: 1014)
     Memory: 21.0M
        CPU: 117ms
     CGroup: /system.slice/php-fpm.service
             ├─28282 "php-fpm: master process (/etc/php-fpm.conf)"
             ├─28303 "php-fpm: pool www"
             ├─28304 "php-fpm: pool www"
             ├─28305 "php-fpm: pool www"
             ├─28307 "php-fpm: pool www"
             └─28308 "php-fpm: pool www"

Jun 10 06:36:55 ip-172-31-11-233.ap-south-1.compute.internal systemd[1]: Starting php-fpm.service - The PHP FastCGI Process Manager...
Jun 10 06:36:55 ip-172-31-11-233.ap-south-1.compute.internal systemd[1]: Started php-fpm.service - The PHP FastCGI Process Manager.
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

7. Creating Website User

The website user `bloguser` was successfully created with a home directory under `/home/bloguser`. A password was assigned to enable secure login and SFTP access for website management.



```
[ec2-user@ip-172-31-11-233 wordpress-project]$ sudo useradd bloguser
[ec2-user@ip-172-31-11-233 wordpress-project]$ sudo passwd bloguser
Changing password for user bloguser.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-11-233 wordpress-project]$ id bloguser
uid=1001(bloguser) gid=1001(bloguser) groups=1001(bloguser)
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

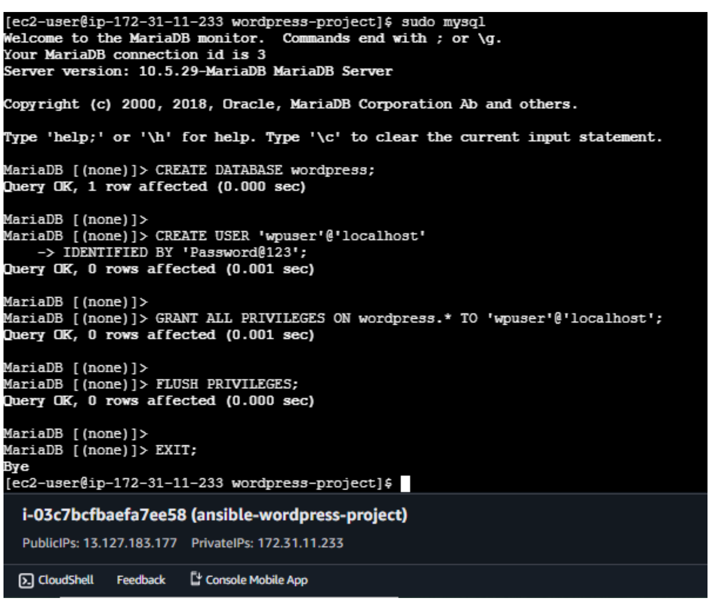

8. Creating Website Structure

The website directory structure was successfully created, and appropriate ownership permissions were assigned to the user for managing website files.

```
[ec2-user@ip-172-31-11-233 wordpress-project]$ sudo mkdir -p /home/bloguser/myblog/public
[ec2-user@ip-172-31-11-233 wordpress-project]$ sudo chown -R bloguser:bloguser /home/bloguser
[ec2-user@ip-172-31-11-233 wordpress-project]$ sudo ls -ld /home/bloguser/myblog/public
drwxr-xr-x. 2 bloguser bloguser 6 Jun 10 06:49 /home/bloguser/myblog/public
[ec2-user@ip-172-31-11-233 wordpress-project]$
```


9. Configuring Database

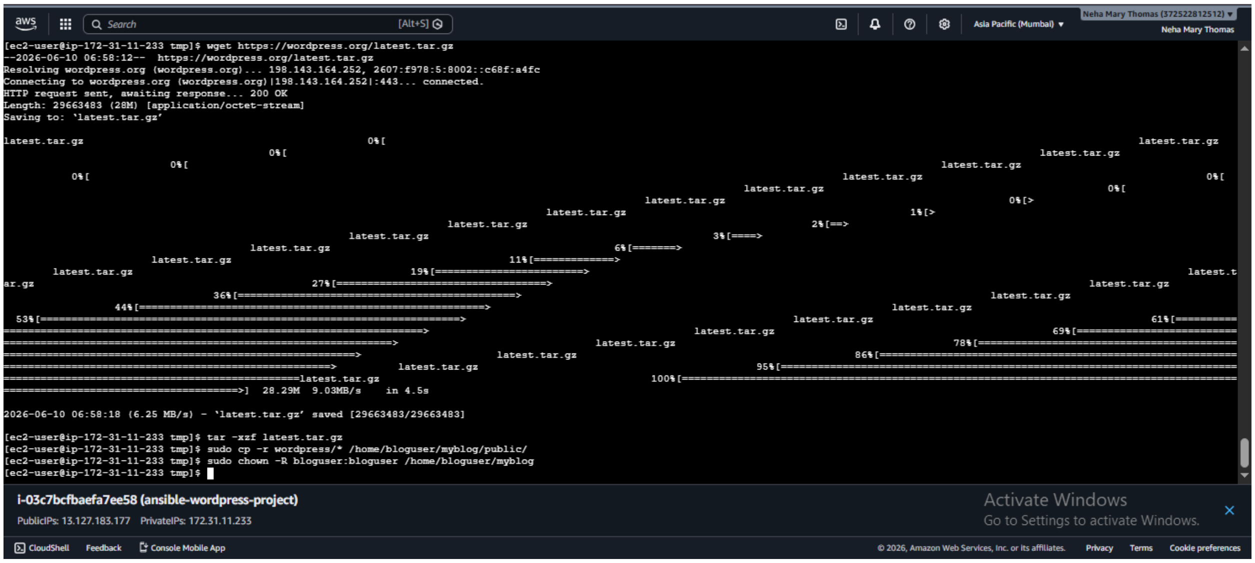
The `wordpress` database was successfully created, and the user `wpuser` was granted the necessary privileges to access and manage the database.



```
[ec2-user@ip-172-31-11-233 wordpress-project]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE wordpress;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]>
MariaDB [(none)]> CREATE USER 'wpuser'@'localhost'
    -> IDENTIFIED BY 'Password@123';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
MariaDB [(none)]> GRANT ALL PRIVILEGES ON wordpress.* TO 'wpuser'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]>
MariaDB [(none)]> EXIT;
Bye
[ec2-user@ip-172-31-11-233 wordpress-project]$
```

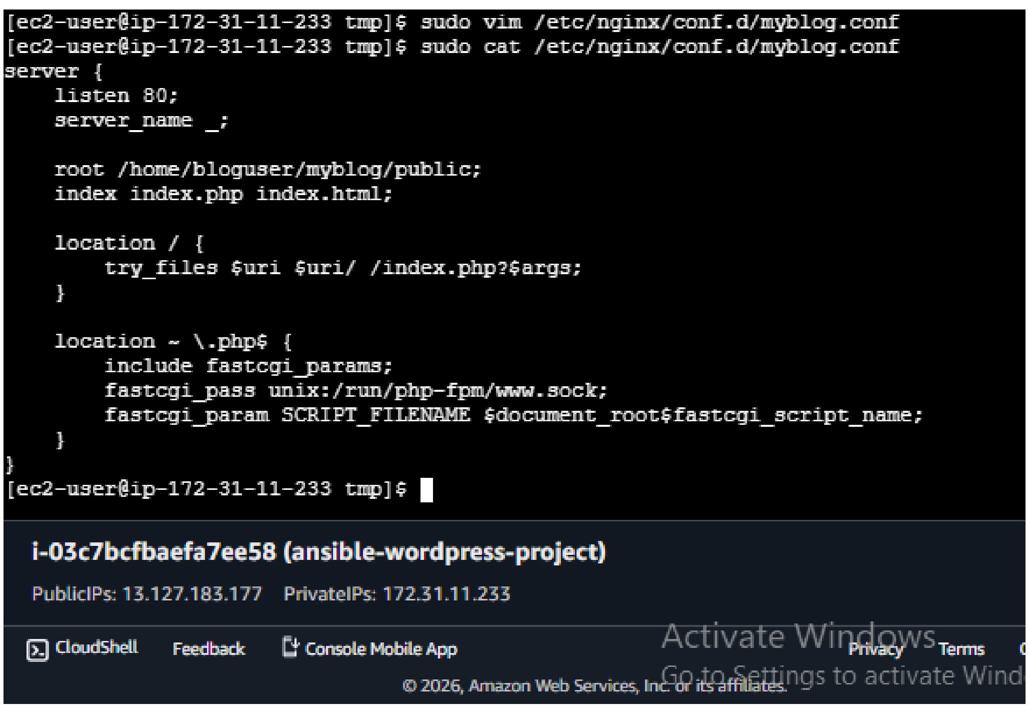
10. Downloading Wordpress

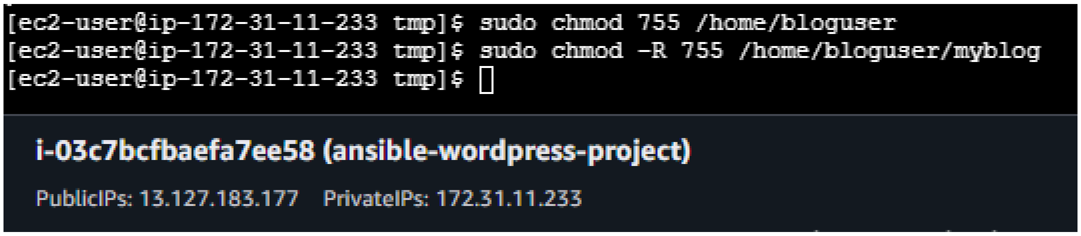
WordPress was successfully installed and configured.



11. Configure and Set permissions for nginx

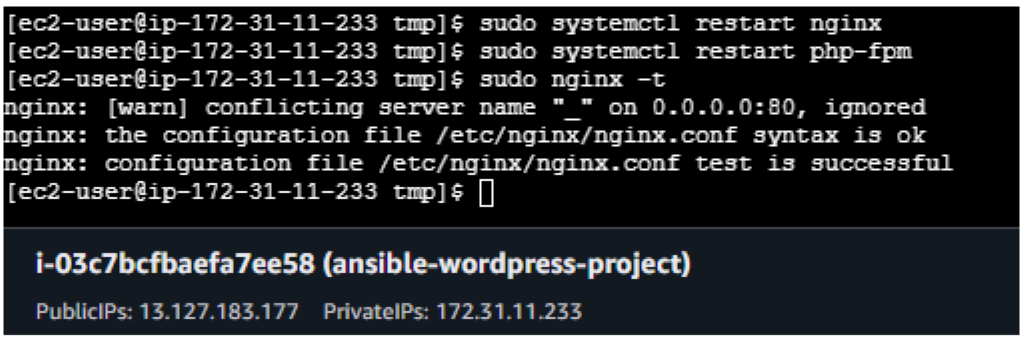
Nginx was successfully configured to host the WordPress website from the custom user directory. Proper file permissions were applied, ensuring secure access to website files.





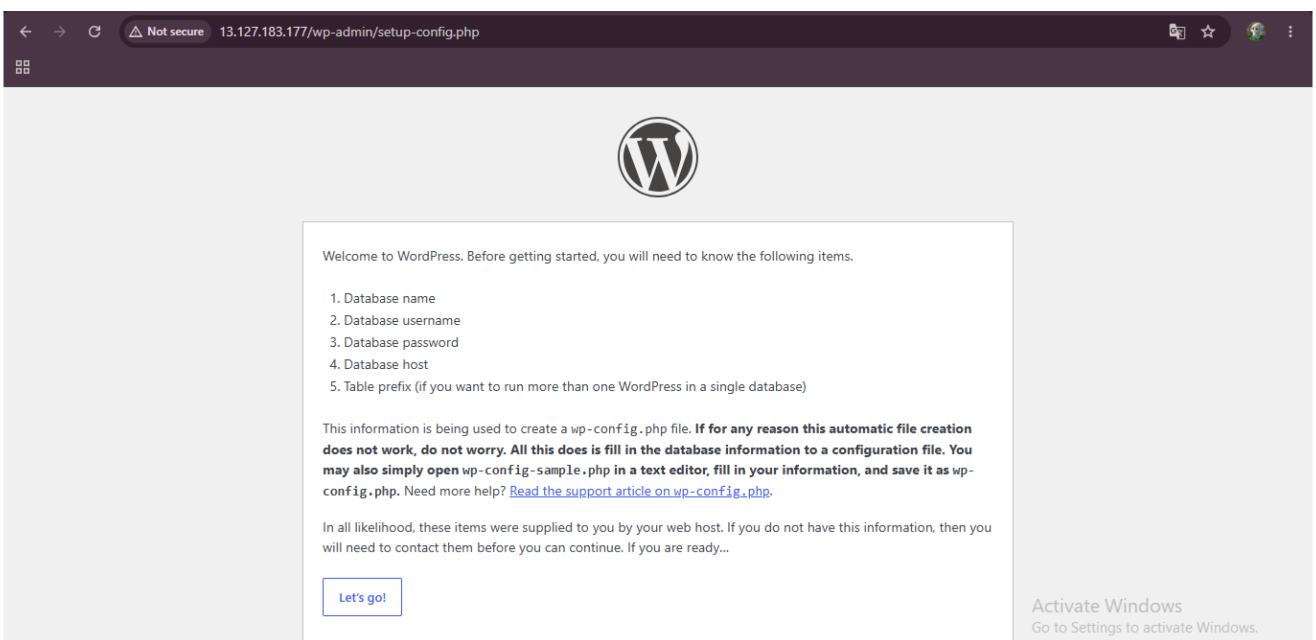
12. Restart Services

The Nginx configuration test completed successfully without any errors, confirming that the configuration file was valid. Both Nginx and PHP-FPM services were restarted successfully, allowing the updated settings to take effect and ensuring proper operation of the WordPress website.



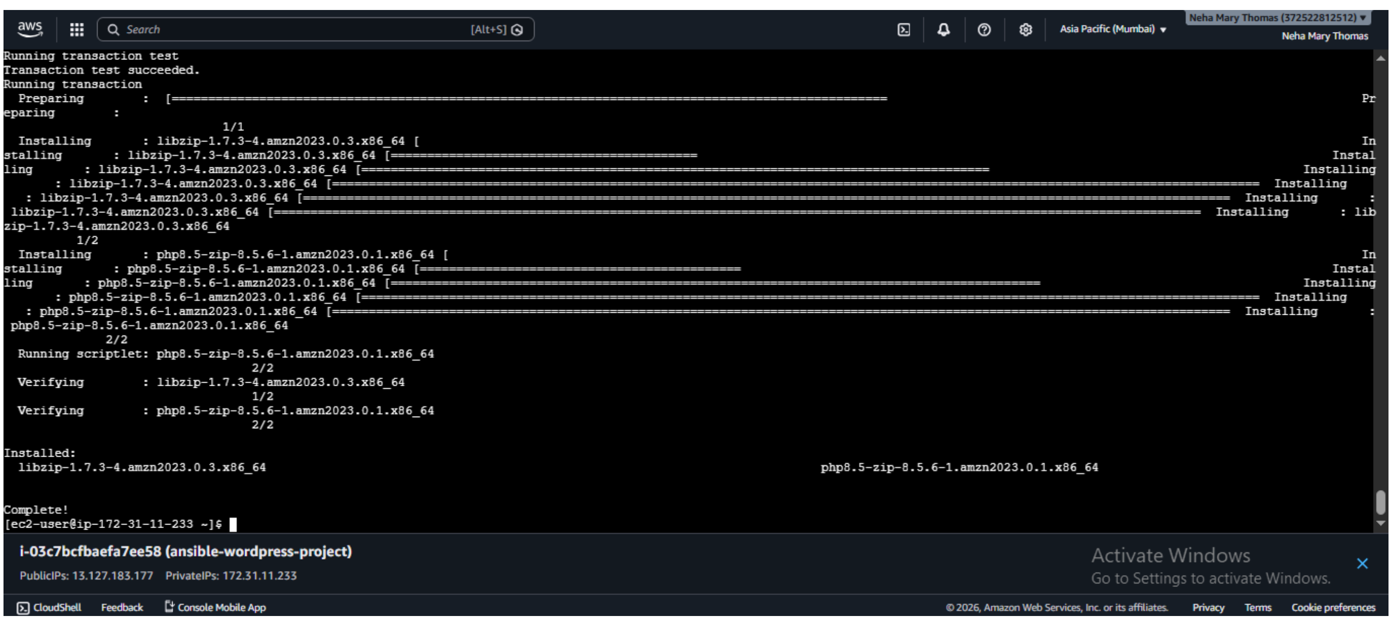
13. Accessing Wordpress In Browser

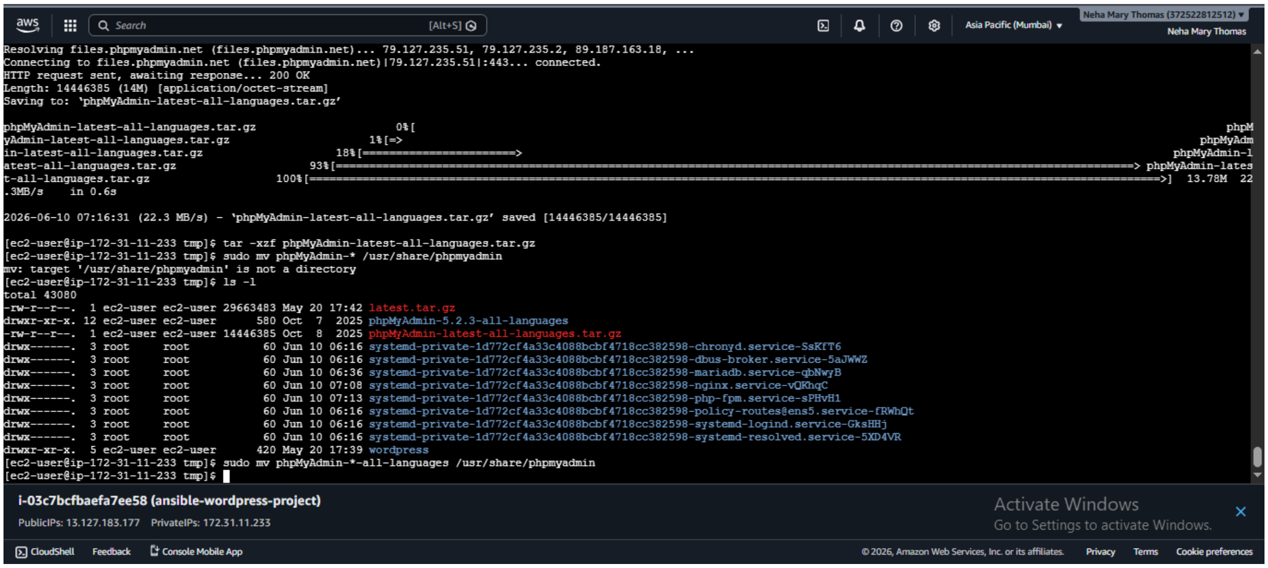
The WordPress installation page was successfully displayed

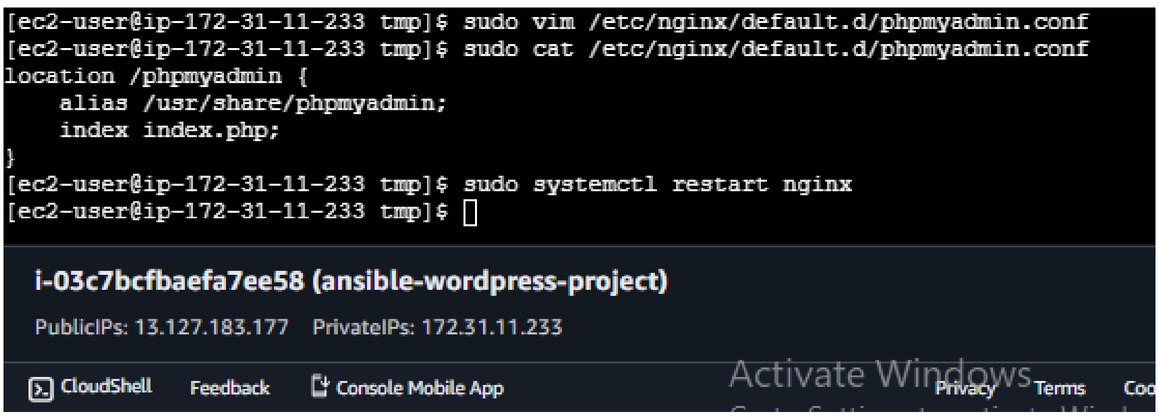


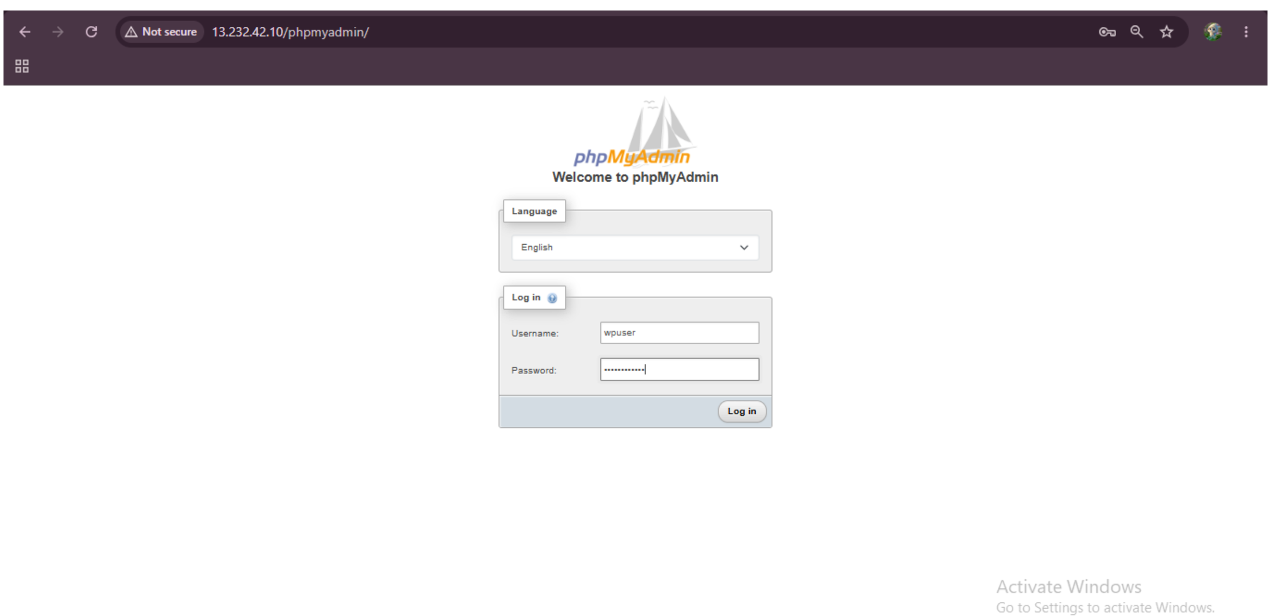
14. Installing phpMyAdmin

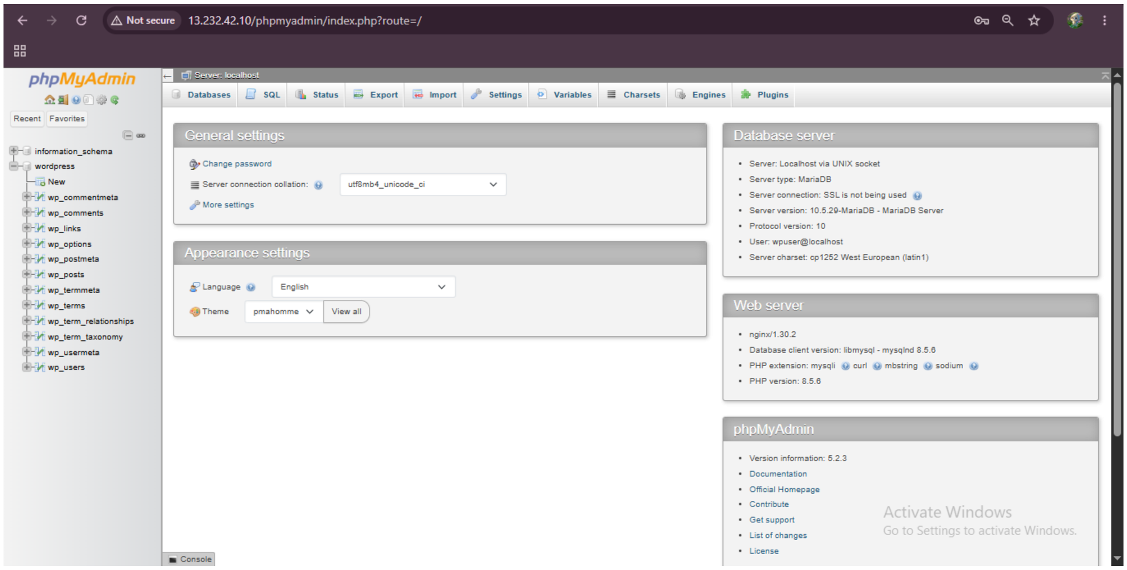
phpMyAdmin was successfully installed and displayed in the web browser. The database management interface became accessible, allowing secure administration of the WordPress database through a graphical web interface.




Search
[Alt+S]
Asia Pacific (Mumbai)
Neha Mary Thomas (372522812512)
Neha Mary Thomas
Resolving files.phpmyadmin.net (files.phpmyadmin.net)... 79.127.235.51, 79.127.235.2, 89.187.163.18, ...
Connecting to files.phpmyadmin.net (files.phpmyadmin.net)|79.127.235.51|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14446385 (14M) [application/octet-stream]
Saving to: 'phpMyAdmin-latest-all-languages.tar.gz'
phpMyAdmin-latest-all-languages.tar.gz 0%[
yAdmin-latest-all-languages.tar.gz 1%[=>
in-latest-all-languages.tar.gz 18%[=======================>
atest-all-languages.tar.gz 93%[
t-all-languages.tar.gz 100%[
.3MB/s in 0.6s
2026-06-10 07:16:31 (22.3 MB/s) - 'phpMyAdmin-latest-all-languages.tar.gz' saved [14446385/14446385]
[ec2-user@ip-172-31-11-233 tmp]$ tar -xzf phpMyAdmin-latest-all-languages.tar.gz
[ec2-user@ip-172-31-11-233 tmp]$ sudo mv phpMyAdmin-* /usr/share/phpmyadmin
mv: target '/usr/share/phpmyadmin' is not a directory
[ec2-user@ip-172-31-11-233 tmp]$ ls -l
total 43080
-rw-r--r--. 1 ec2-user ec2-user 29663483 May 20 17:42 latest.tar.gz
drwxr-xr-x. 12 ec2-user ec2-user 580 Oct 7 2025 phpMyAdmin-5.2.3-all-languages
-rw-r--r--. 1 ec2-user ec2-user 14446385 Oct 8 2025 phpMyAdmin-latest-all-languages.tar.gz
drwx------. 3 root root 60 Jun 10 06:16 systemd-private-1d772cf4a33c4088bcbf4718cc382598-chronyd.service-SsKfT6
drwx------. 3 root root 60 Jun 10 06:16 systemd-private-1d772cf4a33c4088bcbf4718cc382598-dbus-broker.service-5aJWWZ
drwx------. 3 root root 60 Jun 10 06:36 systemd-private-1d772cf4a33c4088bcbf4718cc382598-mariadb.service-qbNwyB
drwx------. 3 root root 60 Jun 10 07:08 systemd-private-1d772cf4a33c4088bcbf4718cc382598-nginx.service-vQKhqC
drwx------. 3 root root 60 Jun 10 07:13 systemd-private-1d772cf4a33c4088bcbf4718cc382598-php-fpm.service-sPHvH1
drwx------. 3 root root 60 Jun 10 06:16 systemd-private-1d772cf4a33c4088bcbf4718cc382598-policy-routes@ens5.service-fRWhQt
drwx------. 3 root root 60 Jun 10 06:16 systemd-private-1d772cf4a33c4088bcbf4718cc382598-systemd-logind.service-GksHHj
drwx------. 3 root root 60 Jun 10 06:16 systemd-private-1d772cf4a33c4088bcbf4718cc382598-systemd-resolved.service-5XD4VR
drwxr-xr-x. 5 ec2-user ec2-user 420 May 20 17:39 wordpress
[ec2-user@ip-172-31-11-233 tmp]$ sudo mv phpMyAdmin-*-all-languages /usr/share/phpmyadmin
[ec2-user@ip-172-31-11-233 tmp]$
i-03c7bcfbaefa7ee58 (ansible-wordpress-project)
PublicIPs: 13.127.183.177 PrivateIPs: 172.31.11.233
Activate Windows
Go to Settings to activate Windows.
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

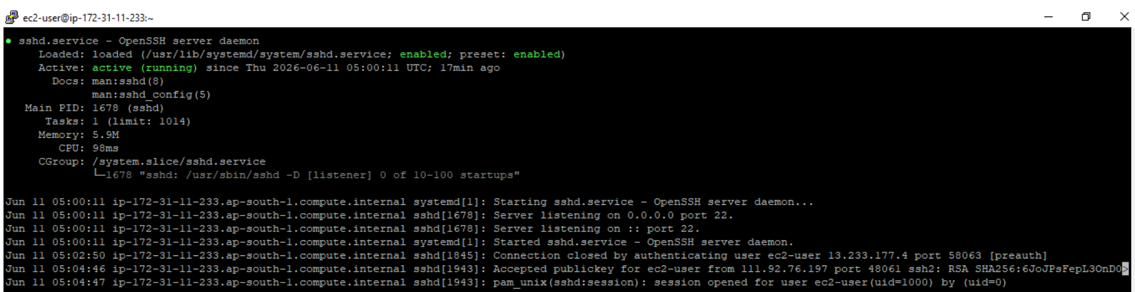

[ec2-user@ip-172-31-11-233 tmp]$ sudo vim /etc/nginx/default.d/phpmyadmin.conf
[ec2-user@ip-172-31-11-233 tmp]$ sudo cat /etc/nginx/default.d/phpmyadmin.conf
location /phpmyadmin {
alias /usr/share/phpmyadmin;
index index.php;
}
[ec2-user@ip-172-31-11-233 tmp]$ sudo systemctl restart nginx
[ec2-user@ip-172-31-11-233 tmp]$
i-03c7bcfbaefa7ee58 (ansible-wordpress-project)
PublicIPs: 13.127.183.177 PrivateIPs: 172.31.11.233
CloudShell
Feedback
Console Mobile App
Activate Windows
Privacy
Terms

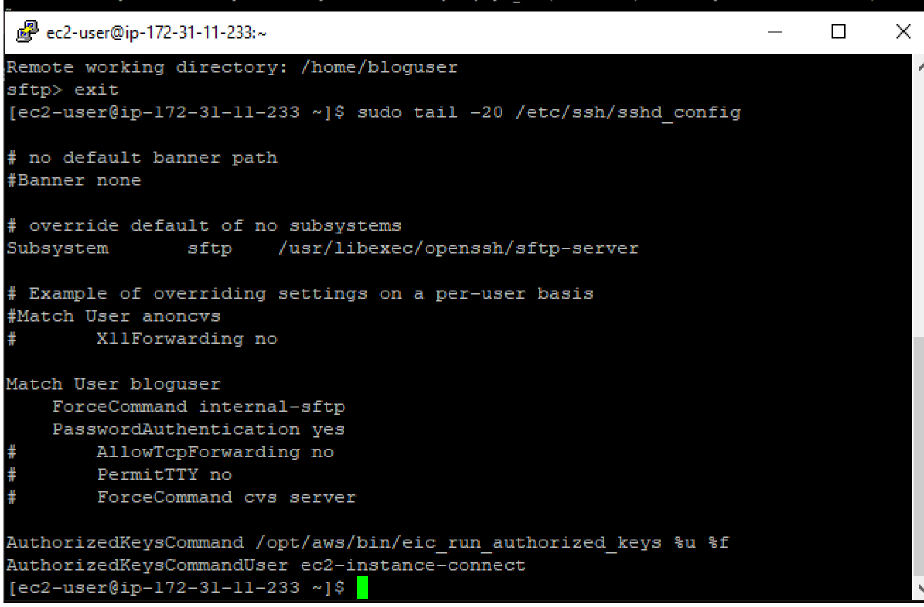

Not secure
13.232.42.10/phpmyadmin/
phpMyAdmin
Welcome to phpMyAdmin
Language
English
Log in
Username:
wpuser
Password:
Log in
Activate Windows
Go to Settings to activate Windows.

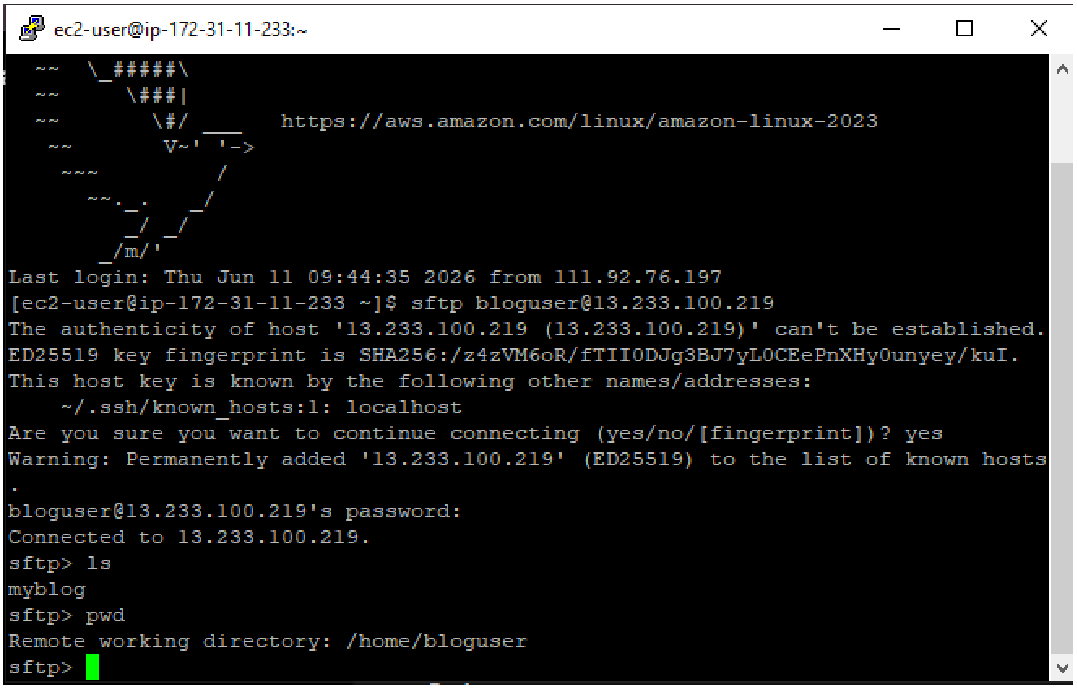


15. Configure SFTP

SFTP was successfully configured and verified. The website user was able to securely access and manage website files through an encrypted file transfer session, fulfilling the secure file management requirement of the project.

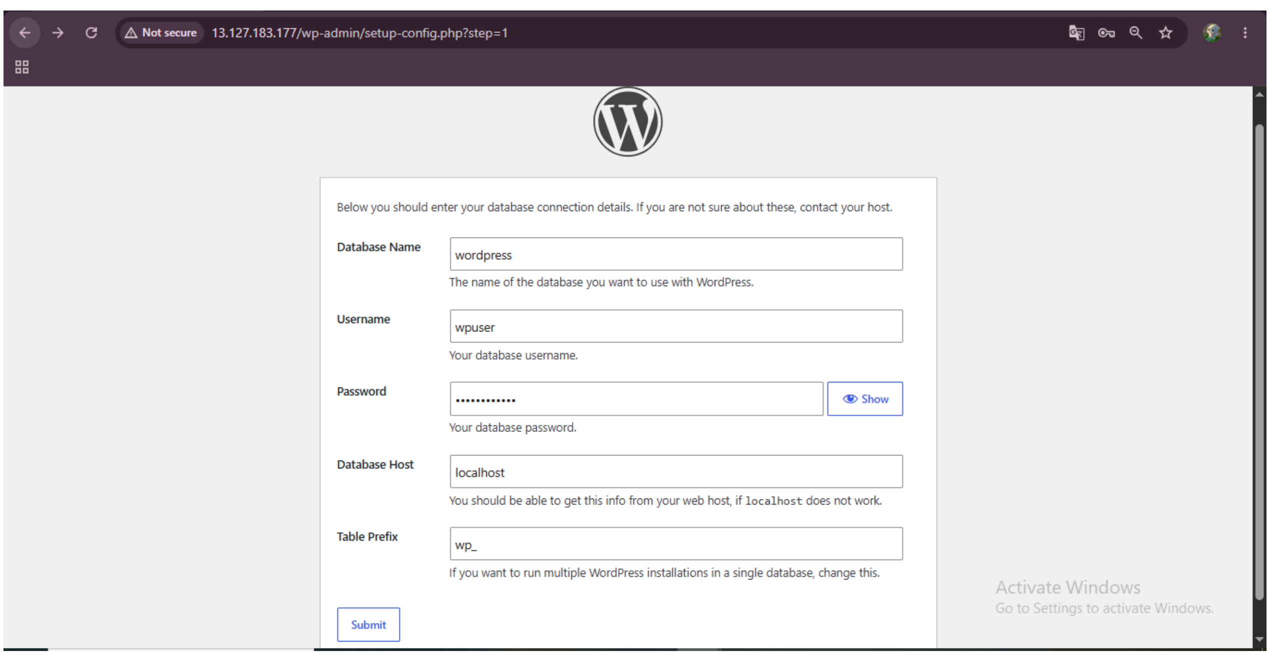


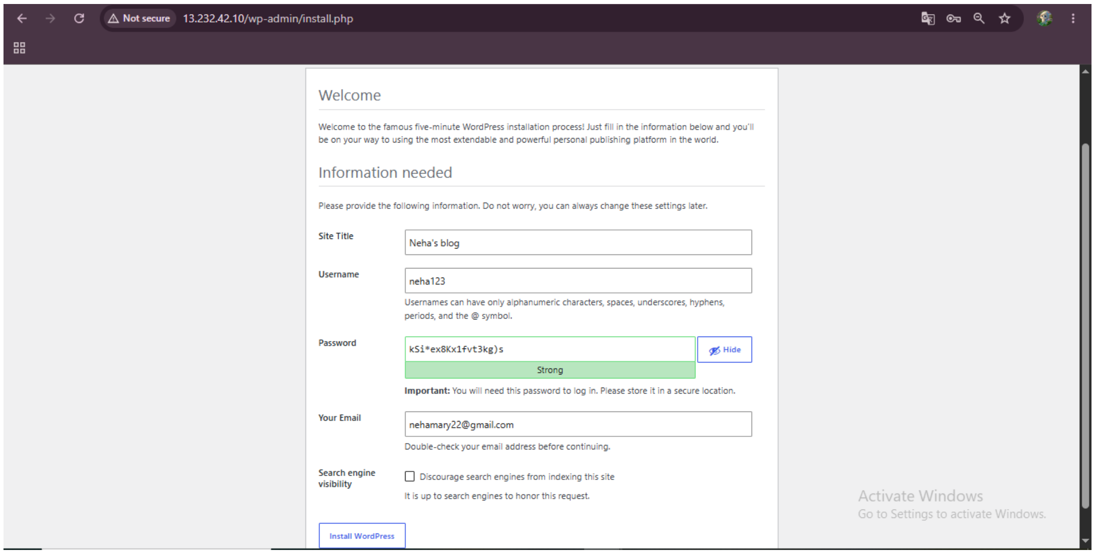




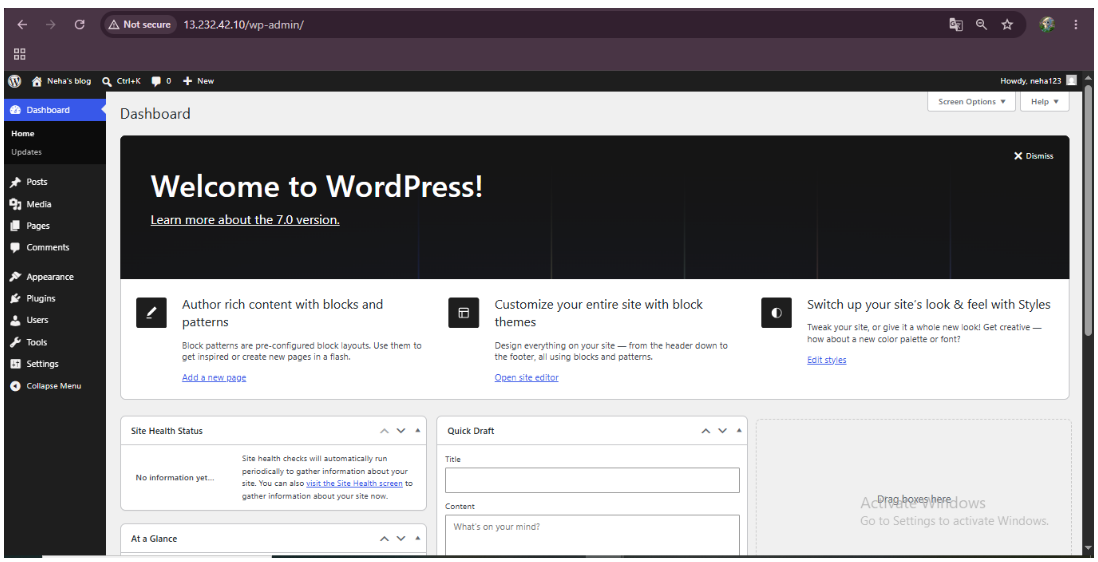
16. Completing Wordpress Setup

The WordPress setup was completed successfully, and the website became fully operational. The published blog content was displayed correctly through the web browser, confirming successful integration of Nginx, PHP-FPM, Database, and WordPress.

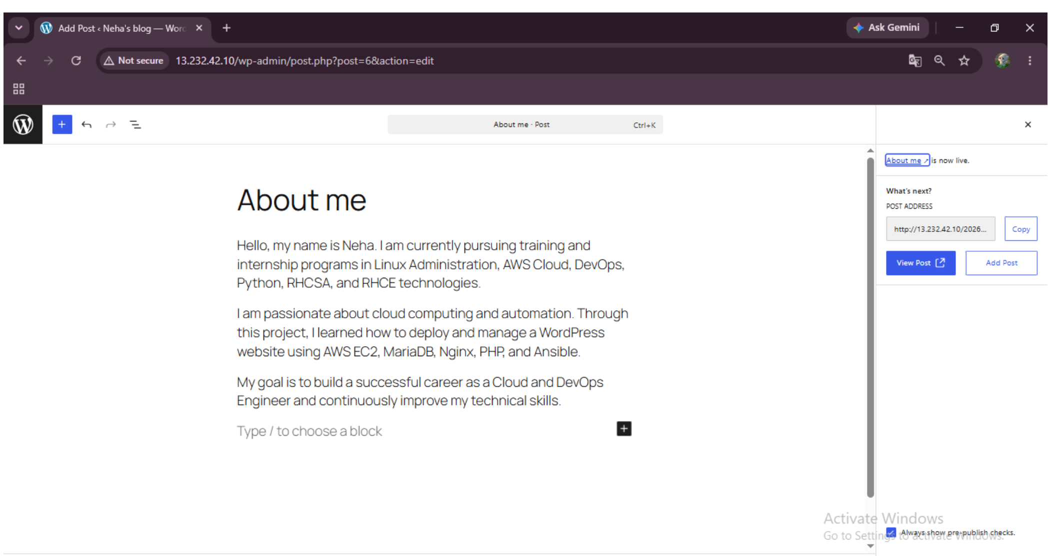




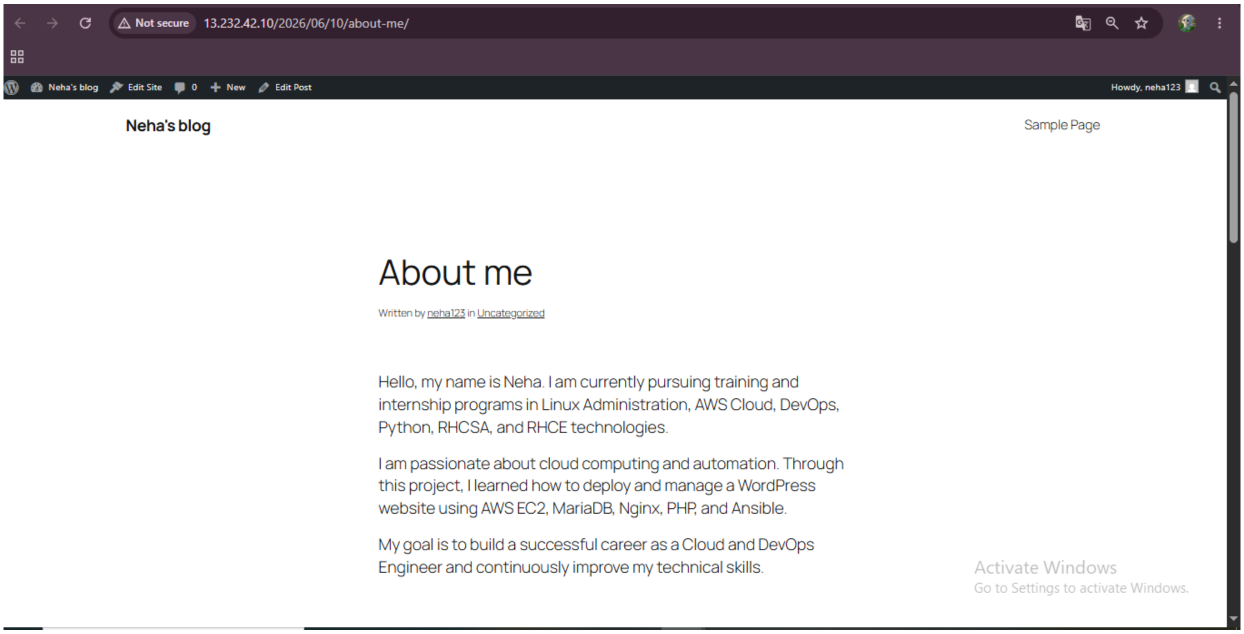
Not secure 13.232.42.10/wp-admin/install.php
Welcome
Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.
Information needed
Please provide the following information. Do not worry, you can always change these settings later.
Site Title
Neha's blog
Username
neha123
Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.
Password
Hide
Strong
Important: You will need this password to log in. Please store it in a secure location.
Your Email
nehamary22@gmail.com
Double-check your email address before continuing.
Search engine visibility
Discourage search engines from indexing this site
It is up to search engines to honor this request.
Install WordPress
Activate Windows
Go to Settings to activate Windows.



Not secure 13.232.42.10/wp-admin/
Neha's blog
Ctrl+K
New
Howdy, neha123
Dashboard
Home
Updates
Posts
Media
Pages
Comments
Appearance
Plugins
Users
Tools
Settings
Collapse Menu
Screen Options
Help
Dismiss
Welcome to WordPress!
Learn more about the 7.0 version.
Author rich content with blocks and patterns
Block patterns are pre-configured block layouts. Use them to get inspired or create new pages in a flash.
Add a new page
Customize your entire site with block themes
Design everything on your site — from the header down to the footer, all using blocks and patterns.
Open site editor
Switch up your site's look & feel with Styles
Tweak your site, or give it a whole new look! Get creative — how about a new color palette or font?
Edit styles
Site Health Status
No information yet…
Site health checks will automatically run periodically to gather information about your site. You can also visit the Site Health screen to gather information about your site now.
Quick Draft
Title
Content
What's on your mind?
At a Glance
Drag boxes here
Activate Windows
Go to Settings to activate Windows.

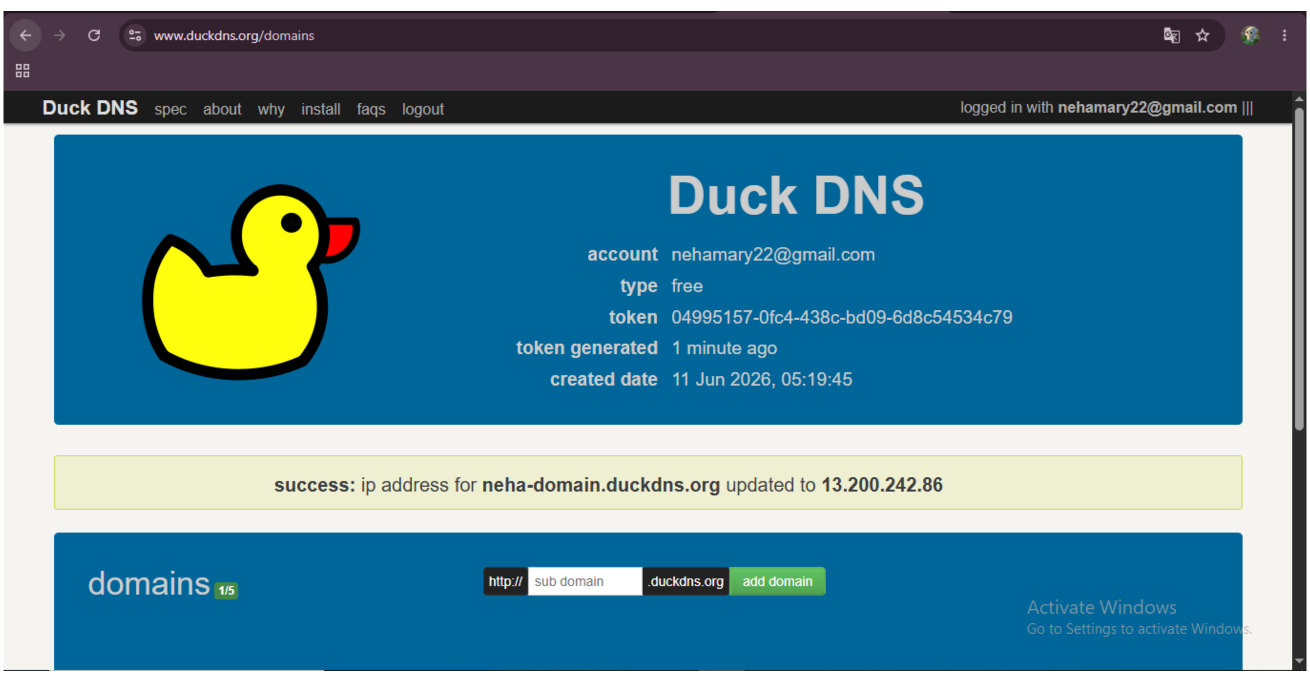


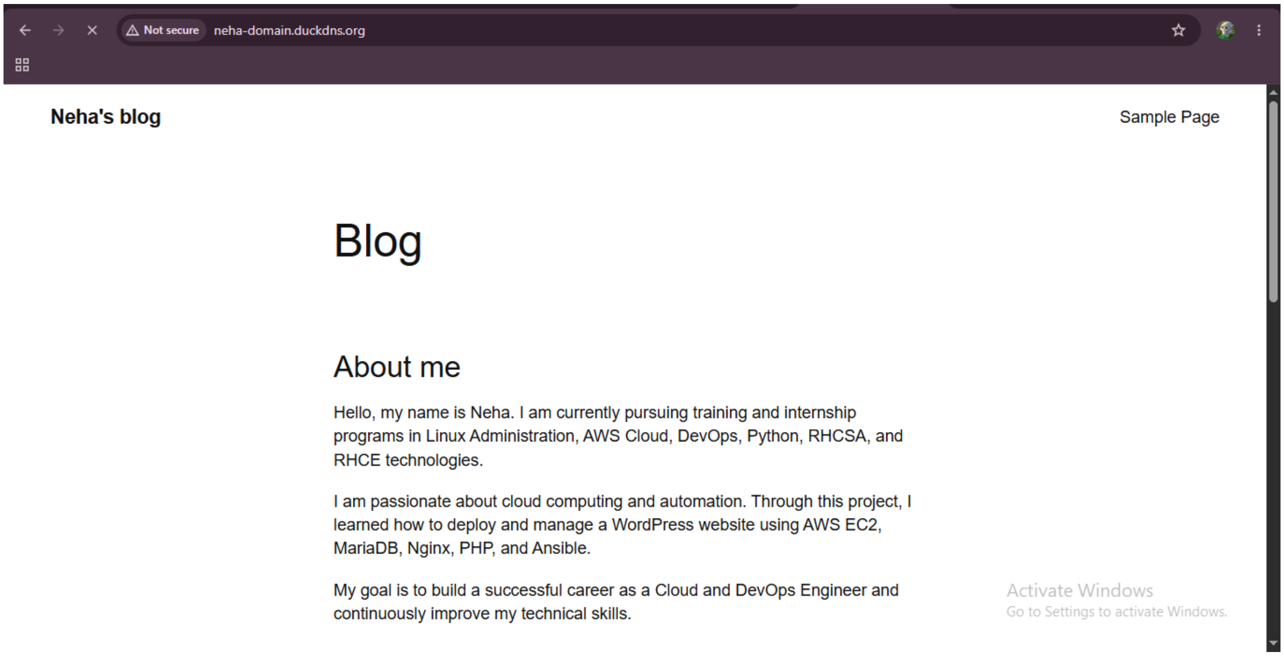
Add Post ‹ Neha's blog — Wor
Ask Gemini
Not secure 13.232.42.10/wp-admin/post.php?post=6&action=edit
About me · Post
Ctrl+K
About me
Hello, my name is Neha. I am currently pursuing training and internship programs in Linux Administration, AWS Cloud, DevOps, Python, RHCSA, and RHCE technologies.
I am passionate about cloud computing and automation. Through this project, I learned how to deploy and manage a WordPress website using AWS EC2, MariaDB, Nginx, PHP, and Ansible.
My goal is to build a successful career as a Cloud and DevOps Engineer and continuously improve my technical skills.
Type / to choose a block
About me is now live.
What's next?
POST ADDRESS
Copy
View Post
Add Post
Always show pre-publish checks.
Activate Windows
Post
Paragraph



17. Domain DNS Configuration

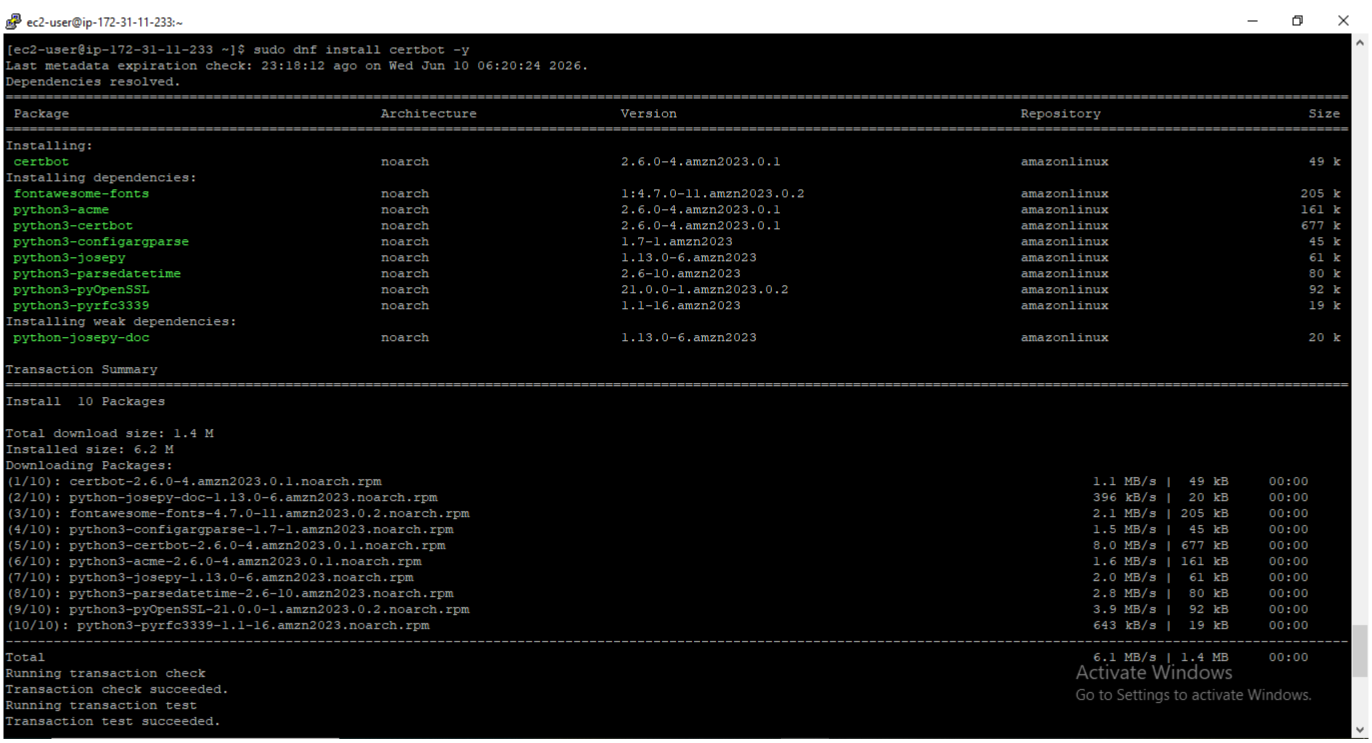
The domain name was successfully mapped to the AWS EC2 instance through DNS configuration. The WordPress website became accessible using the configured domain name, providing a more professional and user-friendly method of accessing the website.





18. Installing Certbot

Certbot was successfully installed and verified on the server. The system was prepared for generating and managing SSL certificates, enabling secure HTTPS access for the hosted WordPress website.



```
[ec2-user@ip-172-31-11-233 ~]$ sudo certbot certonly --standalone --register-unsafely-without-email -d neha-domain.duckdns.org
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.7-June-04-2026.pdf. You must agree in
order to register with the ACME server. Do you agree?
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(Y)es/(N)o: Y
Account registered.
Requesting a certificate for neha-domain.duckdns.org

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/neha-domain.duckdns.org/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/neha-domain.duckdns.org/privkey.pem
This certificate expires on 2026-09-09.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
[ec2-user@ip-172-31-11-233 ~]$
```

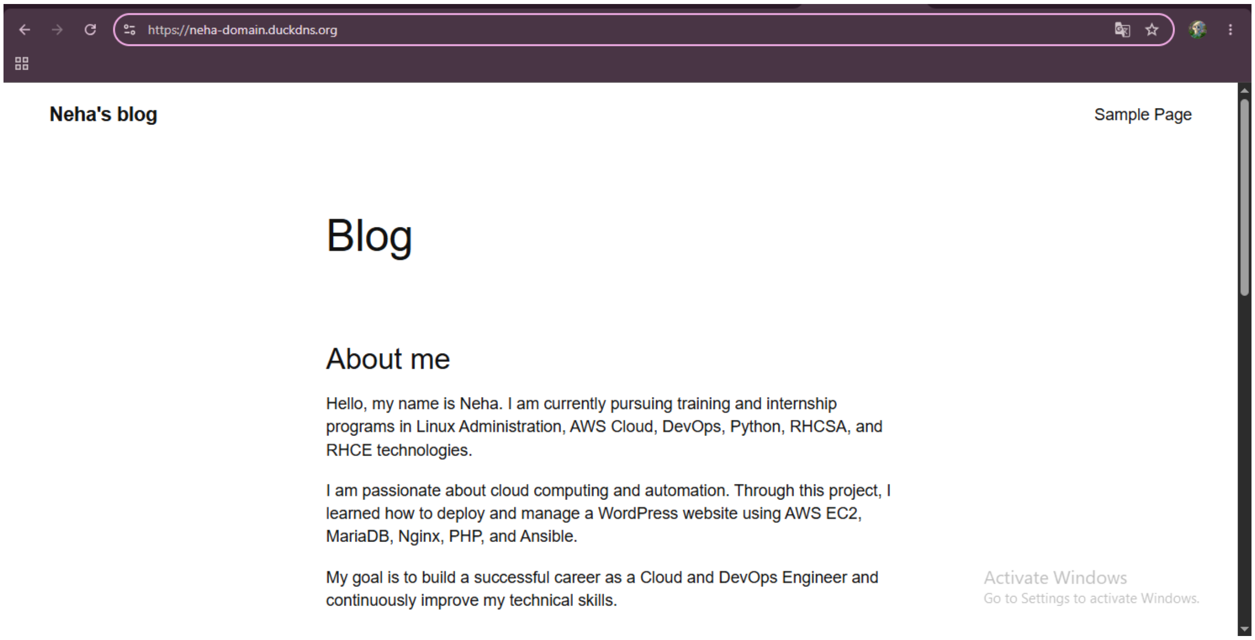
19. Generating SSL Certificate

A valid Let's Encrypt SSL certificate was successfully generated and installed. The website became accessible over HTTPS, ensuring encrypted communication and improved security.

```
[ec2-user@ip-172-31-11-233 ~]$ sudo certbot certificates
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Found the following certs:
  Certificate Name: neha-domain.duckdns.org
    Serial Number: 6d4a4814688285f53c1d2f38478e786cdc8
    Key Type: ECDSA
    Domains: neha-domain.duckdns.org
    Expiry Date: 2026-09-09 04:42:21+00:00 (VALID: 89 days)
    Certificate Path: /etc/letsencrypt/live/neha-domain.duckdns.org/fullchain.pem
    Private Key Path: /etc/letsencrypt/live/neha-domain.duckdns.org/privkey.pem
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
[ec2-user@ip-172-31-11-233 ~]$
```

20. HTTPS Access Verification after SSL Installation

HTTPS access was successfully verified after SSL certificate installation. The website was accessible through a secure encrypted connection, the SSL certificate was active and trusted, and all web traffic was protected using HTTPS.



Conclusion

The project was successfully completed by deploying a fully functional WordPress website on an AWS EC2 instance using Ansible-based automation. All required components, including Nginx, Database, PHP, phpMyAdmin, SFTP access, domain configuration, and SSL security, were configured and verified successfully. A personal blog post was published through WordPress, demonstrating the successful integration of web hosting, database management, secure file transfer, and content management technologies. The project provided practical experience in Linux administration, cloud computing, automation, web server management, and website deployment.

Learning Outcome

This project provided valuable hands-on experience in Linux system administration, AWS cloud services, web hosting, and automation using Ansible. During the implementation process, I learned how to deploy and manage a complete WordPress hosting environment on an Amazon EC2 instance. I gained practical knowledge of configuring Nginx, MySQL, PHP, phpMyAdmin, SFTP, DNS records, and SSL certificates.

One of the most important aspects of this project was troubleshooting various configuration issues related to WordPress, Nginx, SSH/SFTP, and database connectivity. Resolving these issues improved my problem-solving skills and helped me better understand the interaction between different components of a web hosting environment.

Through this project, I also enhanced my understanding of cloud infrastructure, server security, website deployment, and automation concepts. Overall, the project was a valuable learning experience that strengthened my confidence in managing Linux servers and deploying web applications in a real-world environment.

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